PRINCE CHAUHAN

Senior Quantitative Researcher | Options, Volatility & Systematic Trading

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SUMMARY

Options trader turned quant researcher and developer, with nine years on Indian index derivatives (Nifty, Bank Nifty, Sensex). For the last two years I have built and run a fully automated options desk end to end: market data, research, backtesting, execution, risk and live monitoring. A Chartered Accountancy background (CA Final, Group 1) means I care as much about how P&L is measured as how it is made: costs on every leg, no lookahead, and numbers that tie back to a trade ledger.

EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst & Researcher — Dec 2024 - Present

- Built QuantsNerv, the desk's in-house trading platform in Python: live WebSocket tick feeds, real-time IV and Greeks, multi-leg order entry through the DhanHQ API, and browser dashboards for live positions and P&L.
- Run a book of systematic strategies on Nifty, Bank Nifty and Sensex options: short straddles and strangles, opening-range directional selling and buying, and a Bank Nifty constituent basket.
- Combined portfolio backtest, Jan 2021 to Sep 2026 (1,411 sessions of 1-minute data, 0.50% cost charged on every leg): INR 1.41 Cr net on INR 1 Cr capital, Sharpe 3.67, worst drawdown INR 3.8 L (1.7%), 61 of 69 months profitable.
- No strategy goes live without passing the same gate: double-cost stress test, in-sample / out-of-sample / holdout split, parameter stability check and a 1,000-path Monte Carlo of drawdowns.
- Wrote the risk layer every order passes through: margin check before entry, daily loss stop that blocks new trades, forced exit of all intraday positions at 15:20, order slicing to exchange freeze limits, and back-off when the broker rate-limits.
- Run production on two machines, one for live trading and one for research, with a crash supervisor, heartbeat checks and Telegram alerts on every fill, exit and error. Extended the platform to US index options (SPX) through the Alpaca API.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research — Mar 2017 - Nov 2024

- Traded systematic multi-leg option books on Nifty and Bank Nifty (iron condors, calendars, butterflies), rebalancing delta and vega through the day.
- Built the volatility tools behind those books: option price to IV inversion (Newton-Raphson with a Brent fallback), smile smoothing across strikes, and dealer gamma maps showing where hedging flows flip.
- Built a volatility regime filter (k-means on realised volatility and the IV/RV ratio) that cuts short-gamma size when volatility starts to expand.
- Sized every book off exchange SPAN margin on multi-crore capital and kept drawdowns inside desk limits.

Indian Oil Corporation Ltd.; CA Matta & Associates - Treasury; Audit & Tax — 2014 - 2017

- Banking and treasury cash-flow forecasting at IOCL (2016-2017). Statutory audit and corporate tax work at CA Matta & Associates (2014-2016).

DESK RESEARCH & BACKTESTING

- Maintain a backtest library of 13 option strategies on 1-minute NSE/BSE data from 2019 onward. Each run produces the same four outputs: a PDF tear sheet, a VIX-regime breakdown, an Excel report and a leg-by-leg trade ledger, so any number can be traced to the trades behind it.
- Research tracks that feed the live book: India VIX regimes and when short volatility should stand down, IV skew z-scores across strikes, open-interest flow for intraday direction, and cash-futures basis.
- Backtest rules match live trading: entries and exits only on data available at that minute, expiries read from the exchange contract file, freeze-limit slicing, and the 15:20 forced exit.

OPEN-SOURCE RESEARCH CODE

github.com/prince-chauhan-pc/quant-research-portfolio - seven small Python modules, 29 unit tests, CI on every push. Sample data only; no live strategy logic.

- Point-in-time backtester with margin-based lot sizing, per-leg costs, Sharpe / Sortino / Calmar and a standard trade ledger.
- Volatility surface: Black-Scholes IV inversion, SVI fit per expiry, Greeks and a dealer gamma (GEX) profile.
- Pairs and baskets: Engle-Granger cointegration, rolling hedge ratio, mean-reversion half-life, z-score entries.
- Regime filter, a queue-position fill simulator with square-root market impact, a strategy validation gate with Monte Carlo stress, and a PDF tear sheet generator.

SKILLS

- **Quant:** Option pricing and Greeks, IV surfaces (SVI), volatility regimes, cointegration and pairs trading, walk-forward testing, Monte Carlo, VaR, SPAN margin sizing.
- **Code & data:** Python (pandas, NumPy, SciPy), asyncio, WebSockets, Flask, SQLite, DuckDB, PostgreSQL, Parquet, Git.
- **Markets & brokers:** NSE / BSE index options and futures, US index options; DhanHQ and Alpaca APIs.
- **AI tooling:** Use AI coding agents (Claude Code, Gemini) inside written test gates to review code and check backtests for lookahead and cost errors.

EDUCATION & REGISTRATION

- **B.Com (Honours):** Kirori Mal College, University of Delhi, 2011 - 2014. First Division (71%).
- **CA Final:** Institute of Chartered Accountants of India, 2018. Group 1 cleared (78%).
- **Registration:** SEBI Registered Research Analyst. NISM Series-XV.